

SafeLog AI — Architecture Report

Author: Szymon Iwacz · **Stack:** Rails 8.1 monolith, SQLite, Devise, server-rendered ERB · **Demo:** <https://safelog-ai.fly.dev/>

M4 synthesis (four lesson artifacts): M4L2 repo map → M4L3 intake flow research → M4L4 ranked refactors → M4L5 domain distillation + invariant/ACL plans. This report is a concise two-pager built from those artifacts, not a single-prompt dump.

1. System shape (M4L2 — repo map)

SafeLog AI helps engineers debug production incidents from **multiple pasted log sources** without leaking secrets. **Deterministic redaction runs in memory before any persistence or AI call**; raw paste and placeholder mappings exist only for the request lifetime.

```
User → DebuggingCasesController → Intake::ProcessCaseSubmission
      → Redaction::Engine → SQLite (encrypted)
Controller → Analysis::AnalyzeCase → Correlation::ExtractSignals
                                   → Ai::Client (via PromptBuilder) → AiReport
Boundary: Redaction must not import Ai
```

Territory: product logic in `app/services/{intake,redaction,correlation,analysis,ai}/` ; thin controllers; security oracles in `spec/requests/*_security*`. High activity in `context/changes/` is **documentation workflow**, not runtime — a catalog illusion called out in [repo-map.md](#).

Structure: Ruby services form a **DAG** (no cycles). Verified boundary: **redaction ⊥ ai** — constant scan + dependency analysis ([artifact-2-structure.md](#)). E2E layer stays black-box (`depccruise:validate` — 0 violations).

2. Intake flow and refactor direction (M4L3–M4L4)

Researched path: `POST /debugging_cases` → `Intake::CaseSubmission` → `Intake::ProcessCaseSubmission` (transaction, redaction, persist) → show with sanitized evidence and redaction summary ([case-submission-flow-analysis/research.md](#) , ast-grep verified).

Key findings:

- Single production path from paste to persist; no parallel intake shortcuts.
- `redaction_findings.create!` has one call-site — implicit contract between `Redaction::Engine` and persistence.
- Tests prove raw substrings never reach SQLite; AI prompts use placeholders only.

Ranked refactor opportunities ([refactor-opportunities/research.md](#) — exploration only for M4):

Rank	ID	Opportunity	Rationale
1	TD-2	Typed finding persist boundary	Closes implicit hash contract (partially shipped: <code>RedactionFinding.build_from_engine_finding</code>)
2	IMPL-1	Extract persist object from <code>ProcessCaseSubmission</code>	Composes with TD-2; clearer rollback surface
3	TD-5	CRLF normalization in <code>Redaction::Engine</code>	Low blast radius; paste correctness

M4 scope was **research and ranking**, not mandatory production refactor for the Architect badge.

3. Domain model and planned hardening (M4L5)

Ubiquitous language ([01-domain-distillation.md](#)): debugging case, log source, redaction finding, correlation signal, hypothesis report. Subdomains: **Intake & Redaction** (core), **Correlation** (pure extraction), **Analysis & AI** (hypothesis generation behind adapter).

Invariant INV-G1: no diagnostic content persists until in-memory redaction; raw paste and mappings never stored. Enforced today procedurally in `ProcessCaseSubmission` , schema absence of `raw_*` columns, and test oracles — **not** yet by a structural aggregate type.

Plan-only hardening (M4L5 deliverables — implementation optional post-MVP):

`SanitizedCaseDraft` **aggregate guardian** — typed gate before AR persistence ([02-invariant-aggregate-refactor.md](#)).

`HypothesisGenerator` **port + ACL** — isolate provider details; domain depends on port, adapter wraps `AI::Client` ([03-anti-corruption-layer.md](#)).

4. Quality and security (cross-cutting)

Concern	Mechanism
Tests	280 RSpec + system/Playwright; fake AI in CI; security oracles on intake and analyze
CI	<code>bin/ci</code> ↔ GHA parity (RuboCop, Brakeman, bundler-audit, RSpec)
Encryption	Active Record Encryption on sanitized logs, reports, correlation payloads
Access	<code>current_user.debugging_cases.find</code> → 404 cross-user

Reviewer narrative: transient raw intake → encrypted sanitized evidence → scoped ownership → sanitized-only prompts → hypothesis-framed reports with uncertainty notes.

Source artifacts: [repo-map.md](#) (M4L2) · [case-submission-flow-analysis/research.md](#) (M4L3) · [refactor-opportunities/research.md](#) (M4L4) · [context/domain/](#) (M4L5).